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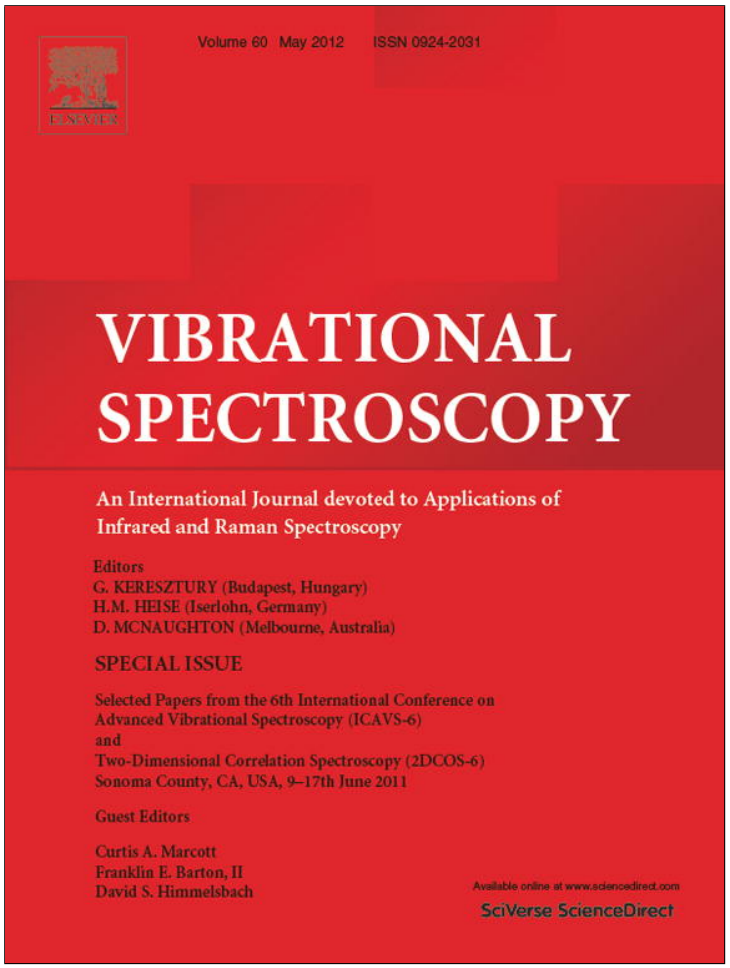

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Study of the pre-melting behavior of polyethylene using Raman spectroscopy combined with two-dimensional (2D) correlation spectroscopy

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ABSTRACT

The thermal behavior of polyethylene (PE) below its melting temperature (T_m) in relation to the density variation was studied by Raman spectroscopy combined with two-dimensional (2D) correlation analysis. The Raman spectra of homo and 1-octene copolymerized PEs were collected by varying the sample temperature from 30 to 100 °C, where pre-melting would occur. The spectral variation upon temperature perturbation was dependent on the density. For a deeper understanding of the thermal behaviors, 2D hetero correlation analysis was performed for the temperature-dependent Raman spectra of homo and 1-octene copolymerized PEs. The hetero-sample correlation analysis revealed that even below T_m, co-monomer side chains excluded from the crystalline lattice of the PE underwent partial fusion, indicating that such a conformational defect was closely related to the pre-melting of PE.

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1. Introduction

Polyethylene (PE) is a semi-crystalline copolymer [1] and its physical properties, such as density, vary with the introduction of side-chain branches. The supermolecular structure consisting of folded-chain crystalline lamellae is present in an amorphous matrix [1,2]. The presence of short branches disrupts the regularity of the molecular structure and lowers the density and crystallinity of PE.

In previous studies, fast and simple determination of the PE density using Raman spectroscopy was demonstrated in combination with a wide area illumination (WAI) scheme to improve the sample representation [3–8]. Furthermore, Park et al. reported that more accurate determination of the density was possible when the Raman spectra of PE samples were acquired at 70 °C [3]. Structural variations induced at the pre-melt stages enhanced Raman spectral selectivity among the samples, thereby resulting in better correlation between the density variation and the corresponding spectral change.

In this study, hetero-sample two dimensional (2D) correlation analysis [9–13] was used to elucidate the polymeric behavior of PE based on the temperature variation and density. The Raman spectra show variations in spectral features depending on the nature of the polymer structure. This variation is also influenced by the change in the density since density is closely related to the polymer structure of PE. Thus, 2D correlation analysis of individual PE samples and hetero-sample 2D correlation analysis become important. A characteristic advantage of the hetero-sample 2D correlation technique is that two different classes of data (two sets of temperature-dependent Raman spectra of PE with different densities) can be directly compared via a hetero correlation technique, which makes it possible to determine the effect of density on the thermal behavior of PE samples. A detailed mathematical description of hetero-sample 2D correlation analysis can be found in relevant Refs. [9–13]. Raman spectra of homo and 1-octene copolymerized PEs were collected by varying the temperature from 30 to 100 °C at 10 °C intervals, and the hetero-sample 2D correlation analysis was performed using both datasets and the pre-melting behavior was examined in detail.

2. Experimental

2.1. PE samples

Two PE samples manufactured from homo polymerization and 1-octene copolymerization were used in this study. The 1-octene copolymerization introduces relatively long side chains in the polymeric structure. The density of each sample was determined using a standard test method as described in ASTM D1505-3 [14]. The density reading of a sample was obtained by maintaining the sample temperature at 23 °C in a constant-temperature bath. The

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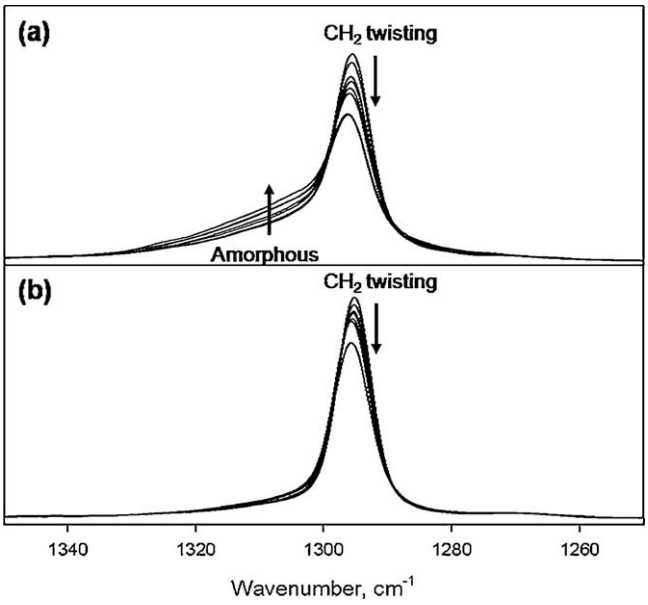

Fig. 1. Temperature-dependent Raman spectra of the (a) 1-octene copolymerized and (b) homo PEs. The arrow indicates the temperature change from 30 to 100 °C.

densities of the homo PE and 1-octene PE were 0.961 and 0.918 g cm^{-3}, respectively.

2.2. Collection of Raman spectra

Raman spectra were collected with the WAI scheme (Kaiser Optical Inc., Ann Arbor, MI, USA) using methods described in our previous paper [3]. An excitation laser (785 nm, diode laser) was magnified to form a circular illumination area with a diameter of 6 mm (illumination area: 28.3 mm^2). The back-scattered Raman radiation was then collected by an array of 50 optical fibers and delivered to a CCD detector. PE pellets were transferred into a circular glass vial (diameter: 9.8 mm). The vial was placed into a metal block with a temperature control of ±0.1 °C. Raman spectra of the PE samples were measured from 30 to 100 °C in 10 °C intervals. Details of this experiment are described elsewhere [3].

3. Results and discussion

3.1. 2D correlation analysis

Fig. 1 shows the temperature-dependent Raman spectra of the (a) 1-octene copolymerized and (b) homo PEs. The 1350–1250 cm^{-1} region was primarily investigated as the spectral variation pertinent to both amorphous and crystalline structures are simultaneously observed in this region. The 1295 cm^{-1} band is assigned to the CH_2 twisting of the crystalline lamellae [15,16].

In the spectra of 1-octene copolymerized PE, the 1295 cm^{-1} band intensity gradually decreased and shifted with increasing temperature, reflecting the change in the mobility of the polymer chain. Alternatively, the 1310 cm^{-1} band corresponding to the amorphous structure, in which the orientation of the polymer chain is disordered, increased with temperature, although the melting temperature (T_m) of PE is generally below 100 °C. The generation of the amorphous structure below T_m can be explained as the partial fusion of the crystalline structure of PE, known as pre-melting. Even well below the T_m, PEs gradually undergo partial fusion of less ordered structure with heating due to the presence of considerably broad distribution of the branches with varying lengths and wide range of crystallite size [16]. The pre-melting phenomenon is fundamentally very similar to the normal melting process, except that it proceeds below T_m.

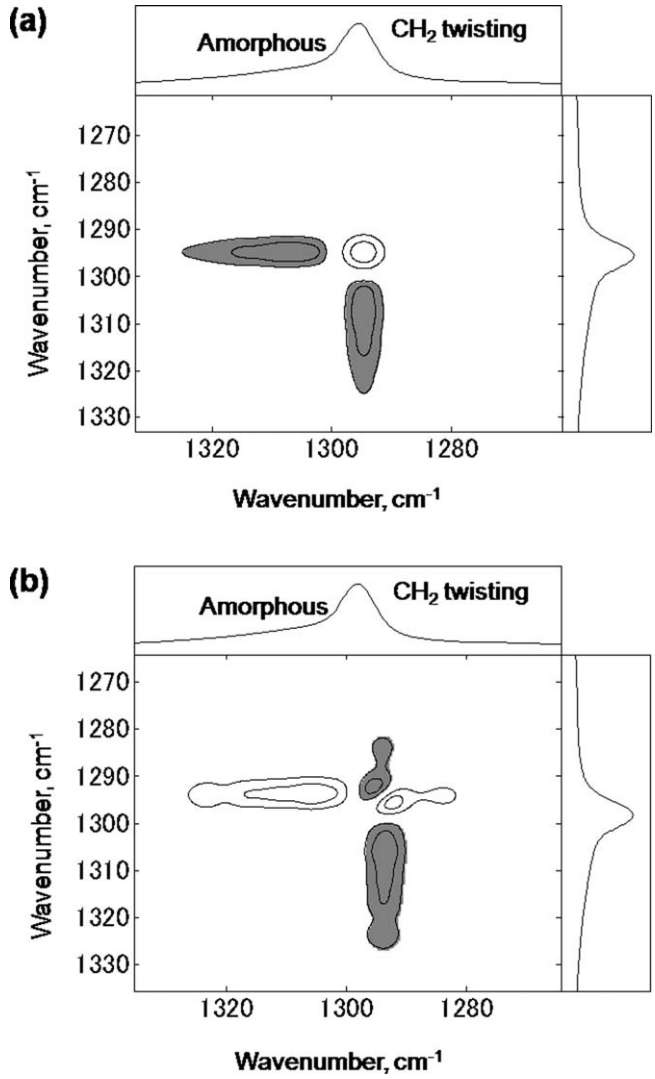

Fig. 2. (a) Synchronous and (b) asynchronous correlation maps generated based on the temperature-dependent Raman spectra of the 1-octene copolymerized PE (low density).

In the homo PE spectrum, the 1295 cm^{-1} band also decreased with increasing temperature; while, the variation in the amorphous band near 1310 cm^{-1} was insignificant compared to that in the spectra of 1-octene copolymerized PE. Therefore, the pre-melting in the homo PE is not dominantly occurring.

The basic idea of generalized 2D correlation spectroscopy is the examination of the synchronicity and asynchronicity of dynamic spectra, which are induced by applied external perturbations, such as temperature and concentration [9–11]. Two-dimensional correlation analysis of the temperature-dependent Raman spectra can be used to elucidate the intrinsic polymer behavior in response to the temperature perturbation, which would be difficult to determine via direct visual examination of the original spectra. Fig. 2 shows the (a) synchronous and (b) asynchronous correlation maps generated based on the temperature-dependent Raman spectra of 1-octene copolymerized PE (low density). The synchronous correlation map has an auto-correlation peak at 1295 cm^{-1}, suggesting the gradual decrease of the crystalline peak. It also has a negative cross peak between 1295 and 1310 cm^{-1}, indicating the gradual decrease and increase of the crystalline

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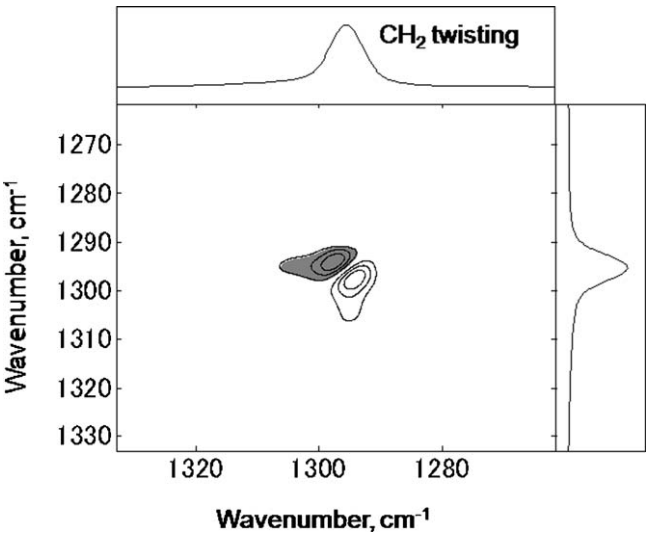

Fig. 3. Asynchronous correlation map generated using the temperature-dependent Raman spectra of the homo PE (high density).

and amorphous components, respectively, during the heating process. The asynchronous correlation map indicates the development of broad cross correlation peaks between 1295 and 1310 cm^{-1}, suggesting that the change in the crystalline component predominantly occurs before the change in the amorphous component. Therefore, the decrease in the crystalline component and the increase in the amorphous component do not occur simultaneously during thermal deformation, indicating the existence of an intermediate between the highly ordered crystalline state and the completely liquid-like amorphous state with partial fusion of less ordered small crystallites, i.e., pre-melting [11,17].

Fig. 3 shows an asynchronous correlation map generated based on the temperature-dependent Raman spectra of the homo PE (high density). Although the synchronous correlation map of the homo PE is not shown, the features are very similar to those of the 1-octene copolymerized PE (refer to Fig. 2(a)). The generation of the cross correlation peak at 1295 and 1298 cm^{-1} in the asynchronous correlation map primarily reflects the gradual intensity decrease and an upward shift of the crystalline band. More importantly, the asynchronous correlation peak between the crystalline and amorphous components observed in Fig. 2 disappears. The absence of the cross correlation peak in turn indicates that variations in these two components occur somewhat simultaneously, implying much less occurrence of partial melting in the homo PE.

3.2. Hetero-sample correlation analysis

In addition to an understanding of the pre-melting behavior of the homo and 1-octene copolymerized PEs, hetero-sample correlation analysis of spectral variations between these two polymers is necessary to gain a significant insight into polymeric behavior upon temperature perturbation. For example, if there is any commonality between the response patterns of system constituents monitored by two different analytical methods under the same perturbation, the correlation should be detected, even between the different classes of spectral signals [12,13]. Fig. 4(a) shows the asynchronous hetero-sample correlation map generated from both temperature-dependent Raman spectra of the homo and 1-octene copolymerized PEs. The synchronous hetero-sample correlation map is not shown since its pattern is largely analogous to that of the synchronous correlation map shown in Fig. 2, revealing that variations in the crystalline and amorphous components occur

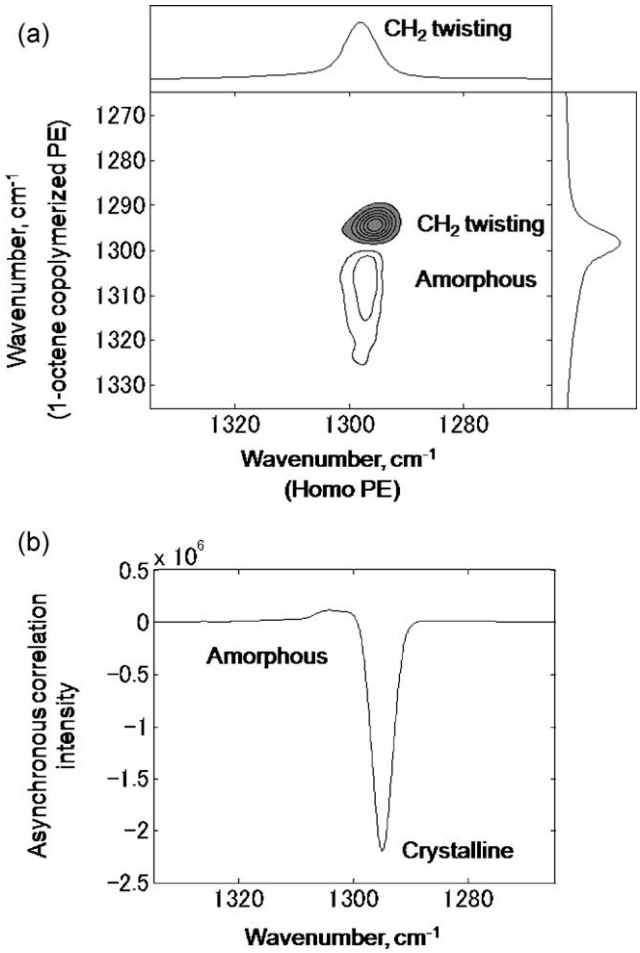

Fig. 4. Asynchronous hetero-sample correlation map generated based on both temperature-dependent Raman spectra of the (a) homo and 1-octene copolymerized PEs and (b) asynchronous correlation intensity along the diagonal direction of the hetero correlation map.

in opposite directions. Therefore, the homo and 1-octene copolymerized PEs undergo very similar structural deformations upon temperature elevation.

Alternatively, a different asynchronous correlation pattern is evident in Fig. 4(a). The asynchronous correlation peak between 1295 (homo PE) and 1310 cm^{-1} (1-octene copolymerized PE) clearly emerges. In the corresponding synchronous correlation map, a negative correlation peak at 1310 cm^{-1} (homo PE) and 1295 cm^{-1} (1-octene copolymerized PE) was observed, suggesting that development of the amorphous structure in the 1-octene copolymerized PE occurs earlier than that in the homo PE. Consequently, the partial melting of 1-octene copolymerized PE occurs at a lower temperature than the homo PE due to the presence of more extensive branches.

It is also useful to examine the asynchronous correlation intensity along the diagonal direction of the hetero correlation map. Fig. 4(b) shows the sliced spectrum along the diagonal direction. The asynchronous correlation spectrum generated from the homo-sample system has no correlation intensity along the diagonal line because of the absence of dissimilarity. Generation of the asynchronous correlation intensity in the hetero-sample system under similar temperature perturbations indicates the different responses of two polymers to the perturbation. For example, a negative asynchronous correlation intensity was clearly observed at 1295 cm^{-1}. The occurrence of the asynchronous correlation intensity in the hetero-sample system indicates these two samples (or

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sample systems) respond differently to an applied perturbation and temperature in this study. A negative asynchronous intensity at 1295 cm^{-1} indicates that the difference in density results in the different thermal dependency of the PE crystalline structure.

Copolymerization typically introduces side chain branches in the PE structure. These branches disturb the crystalline lattice structure and result in a relatively broad crystallite size distribution. Therefore, the structural disordering and rearrangement process occurs, even far below the T_m. Watanabe et al. proposed a conformation model of PE where most of the co-monomer side chains exist outside of the crystalline lattice and thus, the release from the structural constraint by the exclusion eventually leads to the formation of conformational defects [17]. This model is consistent with the typical α-relaxation of the chain motions from 60 to 80 °C observed by NMR spectroscopy [18]. Importantly, the hetero-sample correlation analysis corroborates the development of a greater pre-melting in a PE with longer side chain branches. Such conformational defects are significantly sensitive to the applied temperature because of the exclusion from the systematically folded lamellae, which eventually results in the pre-melting of PE even in a lower temperature range, as confirmed by the hetero-sample correlation analysis.

4. Conclusion

The relationship between the density of PE and the corresponding pre-melting behavior was successfully determined using Raman spectroscopy combined with 2D correlation analysis. The pre-melting of PE predominantly occurs at slightly lower temperatures when more co-monomer side chains are present in the polymer structure. This result is consistent with the previous findings that some portions of the co-monomer side chains are excluded from orthorhombic PE crystals, thus resulting in the partial fusion of the lamellae at a lower temperature range.

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